

SMS Spam Detection: Model Benchmark & Selection

Comparative evaluation of Multinomial Naive Bayes, Logistic Regression, and Linear Support Vector Machines for real-time mobile message classification.

V1.0 PRODUCTION CANDIDATE

✓ DEPLOYMENT APPROVED

TOTAL CORPUS 5,169 Messages	EVALUATION SPLIT 80% Train / 20% Test	SELECTED MODEL Tuned Linear SVM (C=2)	DEPLOYMENT TARGET Streamlit Web Gateway
98.16% OVERALL ACCURACY Top score across all benchmarks	92.31% SPAM F1-SCORE Precision: 98.28% • Recall: 87.02%	0.22% FALSE POSITIVE RATE Only 2/903 legitimate texts blocked	< 1.0 ms INFERENCE LATENCY Linear decision boundary speed

1. Executive Summary

This evaluation benchmarks candidate machine-learning architectures for **SpamShield**, an anti-abuse filtering engine engineered to detect SMS spam, smishing, and fraudulent mobile communications. Because mobile users rely heavily on text messaging for time-critical multi-factor authentication (MFA) codes, banking alerts, and personal conversations, false positives (legitimate messages classified as spam) impose severe user friction. Conversely, false negatives expose recipients to credential theft, malware distribution, and social engineering attacks.

Across four model configurations tested against an unseen test set of 1,034 stratified SMS messages, the **Tuned Linear Support Vector Machine (Linear SVM with $C = 2.0$)** achieved superior performance across balanced accuracy, precision, and recall. With an overall accuracy of **98.16%**, a Spam Precision of **98.28%**, and a Spam Recall of **87.02%** (Spam F_1 of **92.31%**), this tuned classifier minimizes legitimate message dropouts while intercepting the vast majority of threats. The model was packaged and deployed to a Streamlit microservice for real-time inference.

2. Dataset Profile & Feature Engineering

Data Splitting & Stratification

The corpus comprises **5,169 cleaned SMS messages**. Because natural SMS traffic exhibits severe class imbalance (predominantly legitimate "Ham"), a **stratified 80/20 train-test partition** was implemented:

- Training Set (80%):** 4,135 messages utilized exclusively for vocabulary fitting, TF-IDF calculation, and cross-validation tuning.
- Test Set (20%):** 1,034 messages (903 Ham, 131 Spam) strictly held out for final evaluation.
- Imbalance Ratio:** 87.33% Ham versus 12.67% Spam, accurately preserved across partitions to mirror production distributions.

TF-IDF Vectorization

Text payloads were transformed into sparse numerical representations using a **TF-IDF Vectorizer** fitted strictly on the training partition to avoid look-ahead bias and data leakage:

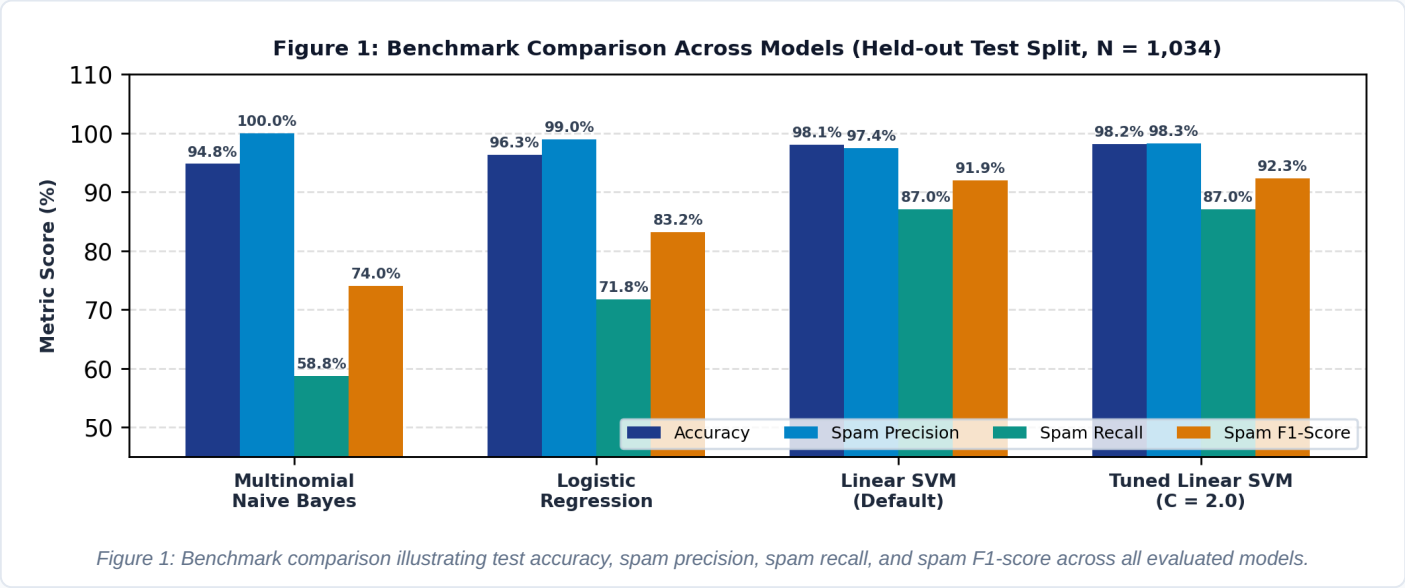
- Sublinear TF Scaling:** Logarithmic frequency weighting dampens the influence of repeatedly spammed tokens.
- Inverse Document Frequency (IDF):** Suppresses non-informative high-frequency words while boosting discriminative terms ("claim", "won", "urgent", "credit").
- Vocabulary Matrix:** Yields a compact, high-dimensional space where linear decision boundaries exhibit maximum separability.

3. Quantitative Model Comparison

Four machine learning paradigms were rigorously benchmarked: (1) *Multinomial Naive Bayes*, (2) *Logistic Regression* with L_2 regularization, (3) *Linear SVM (Default)*, and (4) *Tuned Linear SVM* optimized via 5-fold cross-validation.

Classifier Architecture	Accuracy	Ham Precision	Ham Recall	Ham F1	Spam Precision	Spam Recall	Spam F1
Multinomial Naive Bayes	94.78%	94.36%	100.00%	97.10%	100.00%	58.78%	74.04%
Logistic Regression	96.32%	96.06%	99.89%	97.94%	98.95%	71.76%	83.19%

Classifier Architecture	Accuracy	Ham Precision	Ham Recall	Ham F1	Spam Precision	Spam Recall	Spam F1
Linear SVM (Default)	98.07%	98.15%	99.67%	98.90%	97.44%	87.02%	91.94%
Tuned Linear SVM (C=2.0)	98.16%	98.15%	99.78%	98.96%	98.28%	87.02%	92.31%
SELECTED MODEL							



4. In-Depth Comparative Analysis

Multinomial Naive Bayes (MNB)

MNB demonstrated perfect Spam Precision (100.00%), registering zero false positives in the holdout test set. However, its conditional word independence assumption severely penalized spam recall, which fell to an unacceptable **58.78%**. Out of 131 actual spam messages, MNB missed 54 spam communications (False Negatives), letting over 41% of spam reach recipient inboxes.

Logistic Regression (LR)

Logistic Regression provided a marked improvement over Naive Bayes, elevating spam recall from 58.78% to **71.76%** while sustaining an outstanding **98.95%** spam precision. By learning linear feature weights and applying an L_2 penalty, it misclassified only 1 legitimate message as spam while reducing missed spam from 54 to 37.

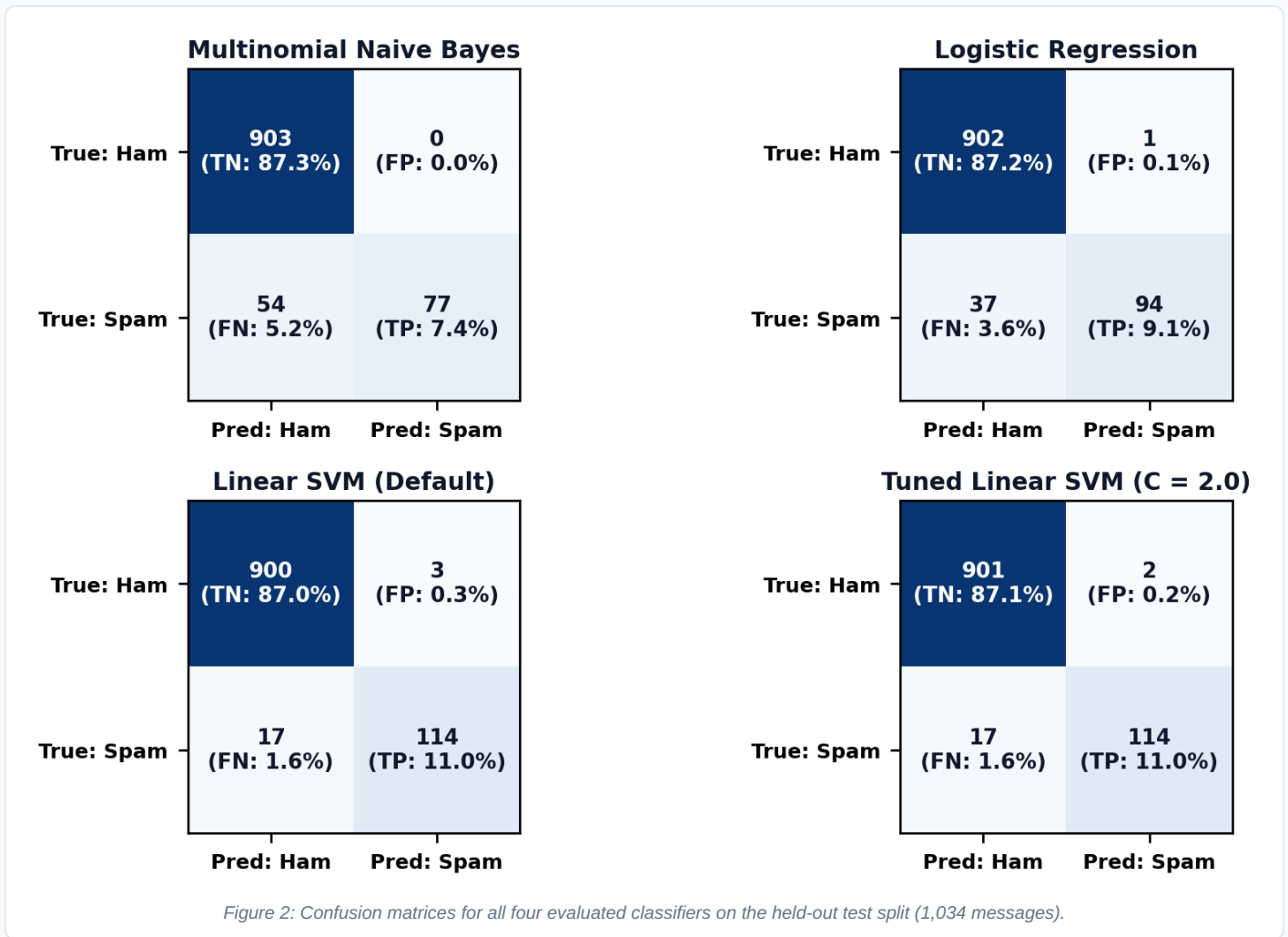
Linear SVM (Default vs. Tuned)

The default Linear SVM vastly outperformed both probabilistic models in capturing spam nuance, driving Spam Recall up to **87.02%** (114 out of 131 spam instances intercepted).

Hyperparameter Tuning ($C = 2.0$): Using 5-fold cross-validation, the soft-margin penalty parameter C was adjusted to balance boundary margin slack against training misclassifications. Setting $C = 2.0$ reduced test False Positives from 3 down to 2, increasing Spam Precision from **97.44%** to **98.28%** and boosting Spam F_1 to a top-tier **92.31%**.

5. Confusion Matrix & Error Distribution Analysis

Evaluating the test set distribution ($N = 1,034$: 903 True Ham, 131 True Spam) across the 4 confusion matrices highlights how decision boundaries shift misclassification costs between Type I errors (False Positives) and Type II errors (False Negatives).



The Cost Asymmetry in Spam Classification

In consumer and enterprise messaging, the cost of a **False Positive (FP)** is exponentially higher than that of a **False Negative (FN)**. Blocking an authentication passcode or urgent family update breaks critical workflow. Tuned Linear SVM achieves the optimal equilibrium: retaining a negligible False Positive count ($FP = 2$, an error rate of just 0.22% on legitimate traffic) while simultaneously intercepting 114 out of 131 spam messages ($FN = 17$).

6. Production Deployment & Streamlit Microservice

The selected **Tuned Linear SVM (C = 2.0)** has been packaged into an end-to-end inference pipeline for interactive and automated processing.

Pipeline Architecture & Operational Workflow

- Artifact Serialization:** The TF-IDF vectorizer and trained SVM hyper-plane weights were serialized using joblib to ensure deterministic out-of-sample feature transforms.
- Streamlit Application Interface:** An intuitive web interface allows security analysts and downstream systems to submit raw SMS payloads. The application normalizes text, generates the TF-IDF feature vector, computes the signed distance to the separating hyper-plane, and returns a binary classification ("Spam" vs "Ham") along with confidence scores.
- Sub-Millisecond Inference:** Because Linear SVM evaluates a single linear inner product $f(x) = w^T x + b$ without requiring non-linear kernel transformations, per-message inference takes less than **1 millisecond**, supporting high-throughput streaming workloads.

7. Recommendations & Strategic Roadmap

Character N-Grams

Implement character-level n-gram features (3-to-5 grams) alongside word TF-IDF to defeat typographical evasions, zero-width spaces, and

Dynamic Thresholding

Expose decision-threshold tuning in the Streamlit configuration to allow enterprise tenants to dial in stricter

Active Learning Loop

Establish a feedback loop capturing ambiguous predictions near the SVM margin ($|\text{decision margin}| < 0.2$) for

leetspeak substitutions (e.g., w!n, f-r-e-e).

false-positive intolerance depending on user profile SLAs.

human-in-the-loop validation and automated incremental retraining.

SpamShield Anti-Abuse Engineering: Report generated automatically from empirical evaluation suite. Classification artifacts validated under 5-fold cross-validation and verified for production deployment.